When AI Becomes the First Reader: From Peer Review to Proof of Contribution

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Abstract

Scholarly work has a new first reader. Before a human peer opens a manuscript, AI systems have often already helped draft it, and they increasingly screen, summarize, rank, and pre-evaluate it; in a measurable share of cases they also help write the review that judges it. This paper argues that the shift exposes a structural flaw in how academic credit is settled. Peer review was designed as the sole settlement mechanism for credit in a world where humans read first. It settles credit at the level of whole papers, through the channel of scarce human reading. Both design assumptions are now failing. Contributions increasingly live in components—datasets, methods, evidence, replications—while credit is still issued per paper, a granularity mismatch. And the human-reading channel is overloaded by volume and fraud, while the AI detectors adopted to protect it are demonstrably unreliable, a channel mismatch. We propose upgrading academic credit from a peer-review stamp to proof of contribution: credit represented as a five-dimensional vector (origination, evidence quality, verification, maintenance, and downstream use), accounted at the level of knowledge components and supported by provenance records, with peer review retained as one settlement mechanism among several. The proposal is an upgrade of peer review, not its abolition. Throughout, we separate established facts, trend judgments, and conditional scenarios, and we state the boundaries of the proposal explicitly. We close with a three-paper design-science research agenda and three testable propositions, one for each of the paper's claims.

Keywords: peer review; academic credit; generative AI; contribution accounting; research provenance; scholarly communication

*[TO FILL: affiliation and contact.] Working paper, first draft. Comments welcome.

1 The first reader has changed

Every scholarly text is read first by someone. For most of the history of modern science, that someone was a human: an editor deciding whether a manuscript merited review, a referee deciding whether it merited publication, a colleague deciding whether it merited citation. This paper starts from a simple observation with large consequences: the first reader of scholarship is increasingly not human.

A note on epistemic status. Because the subject invites overstatement, we label our claims. *Established facts* are documented in the cited sources. *Trend judgments* are our reading of the direction in which documented facts point. *Conditional scenarios* describe what would follow if a trend continues, and are not predictions. *Hypotheses* are claims we propose to test. Claims whose primary source is not yet in our reference list are marked [**TO VERIFY: . . .**]. They illustrate a point but carry no argumentative weight.

1.1 The phenomenon

Consider first the evaluative side of the system. Analysing peer reviews submitted to ICLR 2024 and three other major AI conferences, Liang et al. [**?**] estimate that between 6.5% and 16.9% of review text was substantially modified by large language models (LLMs). An independent study of ICLR 2024 finds that at least 15.8% of reviews were written with AI assistance, and that AI-assisted reviews gave higher scores [**?**]. These are established facts about specific venues, and they coexist with formal prohibitions. In June 2023 the U.S. National Institutes of Health (NIH) barred its reviewers from using generative AI in the peer review process [**?**], and the Australian Research Council adopted a comparable restriction [**TO VERIFY: ARC policy statement**]. The rule says humans read first; the measurements say that, in a non-trivial share of cases, they do not.

The productive side shows the same pattern. Measuring 950,965 papers across arXiv, bioRxiv, and Nature-portfolio journals, Liang et al. [**?**] estimate that by February 2024 about 17.5% of sentences in computer science abstracts had been modified by LLMs. A 2025 survey of 4,946 researchers reports that 63% name a lack of guidelines or training as a key barrier to responsible AI use [**TO VERIFY: Wiley 2025 researcher survey**]. AI is used widely in writing, and the rules that would make that use visible lag behind it.

The publication side is changing too. Some journals and review services are separating dissemination from evaluation: eLife has dropped binary accept/reject decisions, and a publish–review–curate model is being formalized by a coalition of organizations [**TO VERIFY: eLife model; PRC alliance, Oct. 2025**]. Once a preprint is public before any human review,

the first systematic reading of it is likely to be done by the search, recommendation, and summarization systems through which researchers now meet the literature. That last step is a trend judgment, not a measured fact. Its direction is nonetheless consistent with every documented data point above.

1.2 The first reader as an analytical concept

We define the *first reader* of a work as the agent whose reading first converts the text into a judgment that conditions later attention—whether the work is read further, reviewed, cited, funded, or ignored. The first reader is not necessarily the most authoritative reader. It is the one that sets the agenda for everyone else. An editor's desk rejection, a referee's summary, or a search engine's ranking each acts as a filter: it shapes which work reaches later readers and how that work is framed when it arrives.

The concept matters because *whoever reads first shapes what counts*. When the first reader was a human peer, the system of academic credit could reasonably treat human reading as the place where value was recognized. When the first reader is an AI system—summarizing a paper for a reviewer, drafting part of the review, ranking the paper in a discovery feed—recognition is mediated before any human judgment occurs. This mediation is not neutral. A summary picks out some contributions and leaves others out. A ranking rewards some signals and ignores others.

The point is not that AI reading is worse than human reading. In some respects it is faster, broader, and more consistent. The point is that the credit system was not built for it. The analysis also excludes a misreading: we do not claim that AI has replaced human judgment. We claim that it now stands *in front of* human judgment, and that the institutions which convert scholarly work into career-relevant credit have not adjusted.

1.3 The question

The situation can be stated as a single question: *when AI simultaneously helps produce, evaluate, and read scholarship, what should settle academic credit?* The rest of the paper answers it in three steps. Section **??** argues that peer review, as the *sole* settlement mechanism for credit, fails on two counts: granularity and channel. Section **??** proposes an upgrade—proof of contribution—in which credit is a vector, the accounting unit is the knowledge component, and peer review becomes one settlement mechanism among several. Section **??** sets out a research agenda and three testable propositions.

2 The settlement failure

By *settlement* we mean the process by which a scholarly contribution is converted into recognized credit that institutions—hiring committees, promotion panels, funding agencies—can act on. Settlement is distinct from dissemination (making work available) and from evaluation in the broad sense (forming any judgment about it). In the current system, settlement happens almost entirely at one point: a paper passes peer review at a venue, and the resulting publication, attributed to an ordered list of authors, becomes the unit of credit.

Peer review performs this role on the basis of three design assumptions, rarely stated but built into the institution:

(A1) *Human first reading.* A qualified human reads the work before it is credited, and that reading is the source of the credit's legitimacy.

(A2) *Whole-paper granularity.* The paper is the natural unit of contribution, and its author list is an adequate summary of who contributed what.

(A3) *A slow, scarce channel.* Reviewing capacity is limited and reading is slow, which makes the stamp informative: passing through a narrow gate signals quality.

Under these assumptions, making peer review the sole settlement mechanism was a reasonable design. Our claim is that two mismatches have opened between these assumptions and the system they govern.

2.1 The granularity mismatch

The first mismatch concerns the *unit* of credit. Much contemporary research is assembled from components: datasets, software and analysis code, methods and protocols, individual empirical findings, replications. These components have different authors, different lifespans, and different patterns of reuse. A dataset may support dozens of papers; a replication may confirm or overturn a finding years after publication; a widely used piece of code may be maintained by people who appear on none of the papers that depend on it. Settling credit per paper compresses all of this into a single author list, attached to a single document, at a single moment in time.

Generative AI makes the compression harder to sustain. The prevailing norm in journal policy is that AI tools cannot be listed as authors, because authorship entails accountability that a tool cannot bear [**TO VERIFY: publisher and COPE authorship policies**]. The norm is sound, but it shows the limits of the author list as an accounting device. AI can contribute substantially to drafting, coding, or analysis, yet the only available record of that contribution is a free-text disclosure, if one is made at all. In parallel, the author list is

a tradeable commodity. Paper mills—organizations that produce fraudulent manuscripts to order—sell authorship slots, and their output accounts for a large share of recent mass retractions [**?**]. A unit of credit that can neither record a genuine non-human contribution nor resist the sale of fictitious human ones is too coarse to carry the weight placed on it.

2.2 The channel mismatch

The second mismatch concerns the *channel* through which credit is settled. Peer review is a channel of human reading. It has two properties that once made it work: it was assumed to be human (A1), and its scarcity made it selective (A3). Both properties are under pressure.

Scarcity has become overload. More than 10,000 research papers were retracted in 2023, a record, of which more than 8,000 came from a single publisher's clean-up of special issues compromised by organized fraud; the retraction rate more than tripled over the preceding decade [**?**]. These are established facts. Our trend judgment is that they indicate a channel whose capacity to screen has fallen behind the volume and sophistication of what passes through it.

The human property can no longer be assured. Institutions have tried to restore assumption (A1) by banning AI from review [**?**] and by screening submissions with AI-text detectors. The measurement literature suggests the first measure is widely circumvented [**?**, **?**]. The second rests on unreliable instruments. Seven widely used detectors misclassified essays by non-native English writers as AI-generated 61.3% of the time on average, against 5.1% for native writers [**?**]. A systematic test of fourteen detection tools concluded that they are neither accurate nor reliable [**?**]. On the theoretical side, as the distributions of human and machine text converge, the performance of even the best possible detector approaches that of random guessing [**?**]. OpenAI withdrew its own AI-text classifier in July 2023, citing low accuracy [**TO VERIFY: OpenAI announcement**]. A channel whose defining property cannot be verified cannot certify that property.

Institutional responses are also fragmented. A 2024 survey of 200 psychiatry journals found that only 39% had any AI policy, with coverage falling from 56% of top-quartile journals to 20% of bottom-quartile ones; among 60 oncology journals, 96.7% required disclosure of AI use but only 35% specified how the requirement would be enforced [**TO VERIFY: psychiatry and oncology journal policy surveys**]. Disclosure without verification is a statement, not a record.

2.3 Why "sole" is the problem

None of this shows that peer review is worthless. Expert human judgment remains the best available instrument for assessing significance, coherence, and fit with a field's open questions. The failure lies in the word *sole*. Because peer review is the only settlement point, every pressure on the system lands on it: the volume of AI-assisted submissions, the fraud of paper mills, the need to police AI use, and the demand to credit contributions it was never designed to see. A mechanism tuned for one function is being asked to perform five.

The consequences, if the trends persist, can be sketched as conditional scenarios rather than forecasts. *If* AI-assisted submission volume continues to grow faster than reviewer capacity, *then* reviewers will rely more on AI assistance, and the human first reading that legitimizes the stamp will become increasingly nominal. *If* detection remains unreliable, as theory suggests it will [**?**], *then* enforcement of AI bans will fall disproportionately on groups whose writing detectors misjudge, such as non-native English writers [**?**]. *If* credit continues to settle only per paper, *then* incentives will continue to favour producing papers over producing reusable, verified, maintained components. A further risk compounds these: AI tools may expand scientific output without a matching growth in understanding, producing what Messeri and Crockett [**?**] call "illusions of understanding." A credit system that counts papers cannot tell the two apart. None of these scenarios is inevitable. Each follows from keeping a single settlement mechanism under conditions it was not designed for.

3 From peer review to proof of contribution

We propose upgrading academic credit from a peer-review stamp to *proof of contribution*. By this we mean a verifiable record, supported by provenance, stating that a particular agent made a particular kind of contribution to a particular knowledge component, together with a record of how that component has been checked and used. The proposal has three parts: a vector of credit, a change in the accounting unit, and a plurality of settlement mechanisms. We state plainly what it is not: *this is an upgrade of peer review, not its abolition.* Peer review remains, with a narrower and better-defined role.

3.1 Credit as a vector

A peer-review stamp is effectively a scalar: a paper is published or it is not, in a venue of a given rank. We propose to represent credit as a vector with five dimensions, defined in Table **??**.

A vector does three things a scalar cannot. First, it lets different institutions apply

Dimension	What it records	Typical evidence
Origination	Who first proposed a question, idea, design, or artefact	Dated provenance records; contribution statements
Evidence quality	How strong the support for a claim is	Design, data, and analysis assessed by reviewers or standards
Verification	Whether, and by whom, a component has been independently checked	Replications, re-executions of code, audits
Maintenance	Who keeps a component usable, correct, and current over time	Version histories; issue resolution; curation records
Downstream use	How a component is reused by later work	Reuse and derivation records, not only citations

Table 1: The five dimensions of the credit vector.

different weights for different purposes without discarding information. A data infrastructure fund may weight maintenance heavily; a prize for discovery may weight origination. Second, it separates dimensions that AI affects very differently. Generative AI has sharply reduced the cost of producing fluent text, which bears on how origination is attributed. It has done far less to reduce the cost of independent verification or long-term maintenance. A scalar credit system cannot register that shift; a vector can. Third, a vector removes the need to treat AI involvement as a yes-or-no property of a whole document.

Current policy shows why that last point matters. At one extreme, the NIH has stated that applications substantially developed by AI will not be considered the original ideas of the applicant [**?**]: AI involvement disqualifies the whole document on the origination dimension. At the other, an experiment at ICLR 2025 reports that LLM-generated feedback on reviews led 27% of reviewers to revise them, and that the revised reviews were rated higher in quality [**TO VERIFY: ICLR 2025 review-feedback experiment**]: AI involvement improves an evaluative contribution. Both judgments may be correct, because they concern different dimensions. A vector lets an institution record that a component's origination was human, that its text was AI-assisted, and that its verification was carried out by an independent team, without forcing these facts into a single verdict.

3.2 The knowledge component as the unit of account

The second move is to shift the unit of account from the paper to the *knowledge component*. We use the term for any separately identifiable, reusable unit of scholarly work that can be attributed, checked, and built on. We distinguish four kinds: *data* (datasets and their documentation), *methods* (protocols, models, and code), *evidence* (specific claims together

with the analyses that support them), and *replications* (independent attempts to reproduce or extend a claim). The paper does not disappear. It becomes a narrative that assembles components, motivates them, and interprets them, and it can itself be credited for those functions.

Contribution accounting at the component level involves four operations, described here in neutral terms rather than as a specific system design:

(i) *Identification.* Each component receives a persistent identifier and a description sufficient for others to locate and reuse it.

(ii) *Provenance.* Each component carries a record of its derivation: which earlier components it builds on, which agents—human researchers and, where relevant, AI tools—performed which operations, and when. Provenance replaces after-the-fact disclosure with a record made as the work is done.

(iii) *Verification.* Claims about a component (that the code runs, that the analysis reproduces, that the finding replicates) are attached to it as separate, attributable contributions.

(iv) *Crediting.* Credit to persons is computed by aggregating their contributions across components, dimension by dimension, rather than inferred from position in an author list.

Two features of this design bear directly on the failures described in Section **??**. Provenance gives AI involvement a place in the record that is neither forbidden nor hidden, which removes the dependence on unreliable detection. And because verification and reuse are recorded as contributions, the incentives that currently favour producing new papers over checking and maintaining existing work are, at least in principle, rebalanced.

3.3 Plural settlement mechanisms

The third move is to replace a single settlement point with several, each matched to the dimensions it can assess well:

- *Peer review* settles judgments of significance, coherence, and contribution to a field's open questions—the dimensions where expert human reading has no substitute. It also continues to assess evidence quality.
- *Verification settlement* credits replications, re-executions, and audits as contributions in their own right.
- *Use-based settlement* credits components according to recorded reuse and derivation, not only citation.

- *Maintenance settlement* credits the continuing work of keeping components correct and usable.

In this arrangement AI systems can legitimately act as first readers—screening, summarizing, checking consistency—provided their role is recorded in the provenance of the evaluation. Human peer review then concentrates on the judgments that most need it, instead of absorbing every pressure on the system.

3.4 Boundaries of the proposal

The proposal has limits, and we state them so that it is not read as claiming more than it does.

It does not abolish peer review. It narrows peer review's role and relieves it of functions it performs poorly.

It does not hand evaluation to AI. Recording AI's role as a first reader is different from endorsing AI as a final judge. Accountability for every contribution remains with identifiable humans.

It is not a metric to be maximized. Any measure that becomes a target invites gaming, and a credit vector is no exception. Downstream use is especially exposed. A 2026 study of 41.3 million papers reports that AI tools expand individual researchers' impact while narrowing science's collective topical focus by 4.63% [**TO VERIFY: Nature 2026 study of 41.3M papers**]. If that finding holds, a credit system that over-weighted use would reward convergence and deepen the monocultures that Messeri and Crockett [**?**] warn of. The vector is meant to inform judgment, not to replace it.

It is not tied to a particular technology. Identification, provenance, and verification can be implemented in many ways. The argument concerns what should be recorded and credited, not the infrastructure that records it.

It has costs. Component-level accounting adds overhead, may disadvantage researchers with fewer resources for documentation, and raises questions of privacy for unpublished work. The bias of AI detectors against non-native writers [**?**] is a warning that measurement instruments can encode inequity. Any implementation must be evaluated for such effects before it is relied on.

4 Research agenda

The argument so far is a position, not a demonstration. Turning it into knowledge requires a sequence of studies. We frame that sequence as design-science research: building and

evaluating artefacts that address a recognized problem, with the artefact and the evaluation as the main sources of knowledge [**?**].

4.1 A three-paper sequence

1. *Position* (this paper). It identifies the problem—a sole settlement mechanism mismatched to an AI-mediated system in granularity and channel—and states the objectives a solution must meet, corresponding to the problem-identification and objective-definition stages of the design-science research process [**?**].

2. *System.* A working artefact that implements component-level contribution accounting for a bounded domain, with identification, provenance, verification, and crediting operations, demonstrated on real research workflows that include AI tools.

3. *Design-science evaluation.* A structured assessment of the artefact against the stated objectives, and the derivation of design principles that can be carried to other domains, presented so that both the artefact and the principles are recognizable as contributions [**?**].

4.2 Three testable propositions

Each of the paper's claims yields a proposition that later work can test. They are hypotheses, not findings.

P1 (first reader).
In fields where AI-text measurement is feasible, the share of manuscripts that receive an AI-mediated assessment—screening, summarization, or review drafting—before the first evaluative human reading is substantial and rising over time. This can be tested by extending distributional measurement methods [**?**, **?**] and by auditing editorial and review workflows.

P2 (settlement failure).
Credit signals settled solely through whole-paper peer review predict later verification outcomes—successful replication, correction, or retraction—less well than signals derived from component-level records of evidence and verification. This can be tested retrospectively on corpora that link publications to replication and retraction data.

P3 (proof of contribution).
Evaluators given vectorized, component-level contribution records make credit decisions that are more consistent across evaluators, and better aligned with independently verified

contributions, than evaluators given paper-level records alone, at an added cost they judge acceptable. This can be tested in controlled evaluation experiments using the artefact from the second paper.

4.3 A call to the scholarly community

The institutions that settle academic credit—publishers, funders, research-evaluation bodies, and the communities that write their rules—are currently responding to AI mainly through prohibition and disclosure. Both responses assume that the old settlement mechanism can be protected by policing its boundary. The evidence reviewed here suggests that the boundary cannot be reliably policed and that the mechanism behind it is overloaded. We invite these communities to treat the moment as a design problem: to experiment with recording contributions at the level where they actually occur, to give verification and maintenance the credit they deserve, and to let peer review do what it does best. AI has already become the first reader of scholarship. The question is whether the credit system will continue to rely on a single gate, or will learn to account for contributions directly.

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